

NIRMAL MADHUNALA

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Portfolio

LinkedIn

GitHub

Technical Summary

Software Engineering student with experience in building scalable applications, cloud-based solutions (AWS), and data-driven and scalable systems. Strong problem-solving abilities with a focus on innovation, product thinking, and designing applications that address real consumer and enterprise challenges. Skilled in Python, SQL, algorithms, analytics and testing. Collaborates effectively in cross-functional and client-facing teams to deliver high-quality software aligned with business processes and growth objectives.

Projects

OneBox – Real-Time Email Aggregator with AI

Node.js, TypeScript, React, Elasticsearch, Qdrant, Docker

- Designed a **scalable cloud-ready email aggregation platform** handling **100+ emails/day** with real-time IMAP IDLE sync and **98.5% data consistency**
- Implemented AI-based email categorization achieving **98–99% classification accuracy** using Groq Llama 3.3 70B with RAG-powered smart reply suggestions
- Built full-text semantic search across **20k+ indexed emails** with **sub-100ms latency** using Elasticsearch + Qdrant
- Improved system throughput by **40%** via analytics-driven performance tuning and optimized async pipelines
- Containerized **5+ microservices** using Docker, reducing environment setup time by **60%**

KLH MedCare – Healthcare Management Platform

React.js, Material-UI, Django, DRF, MySQL

- Built a **modular full-stack hospital management system** supporting **7+ user roles** and digitizing **10+ core workflows** including patient records, treatments, bed allocation, and inventory with real-time monitoring dashboards
- Implemented **enterprise-grade security** with token authentication, OTP reset, rate limiting, and account lockout, reducing unauthorized access risk by **90%**
- Developed **25+ REST APIs** with optimized MySQL queries, improving data retrieval speed by **45%** and ensuring scalable concurrent request handling

Stock Market Analysis System

FastAPI, MongoDB, MySQL, Pandas, React.js

- Engineered a real-time stock analytics platform processing **1M+ historical market records** with automated data ingestion pipelines
- Built interactive dashboards with **7D–5Y historical analysis**, fuzzy search (approximate string matching), and CSV export, reducing manual analysis effort by **50%**
- Optimized FastAPI endpoints using caching and async I/O, achieving **35% faster API response times**
- Designed a **SQL + NoSQL hybrid data architecture** for structured transactions and high-volume analytical queries

AI-Powered Travel Assistant Web App

Flask, LangChain, Groq

- Developed an NLP-driven travel assistant generating **personalized itineraries in i2 seconds** from natural language queries
- Integrated Groq + LangChain + SerpAPI to fetch and structure **real-time flight and hotel data from multiple sources**
- Applied clean architecture principles, reducing code redundancy by **30%** and improving feature scalability

Skills

- Programming:** Python, Java, C, JavaScript, Node.js, React, Express.js, Flask, Django
- CS Fundamentals:** Data Structures and Algorithms, Operating Systems, and Object-Oriented Programming (OOP)
- Databases:** MongoDB, MySQL, PostgreSQL, SQL, Query Optimization, Indexing, Optimization
- Cloud Platforms:** AWS (EC2, S3), MongoDB Atlas
- Tools:** Git, GitHub, Docker, Postman, REST APIs, Unit Testing, CI/CD Pipelines

Certifications

- MongoDB Associate Database Administrator** – MongoDB University (2026)
- AWS Certified Cloud Practitioner** – Amazon Web Services (2025)
- Advanced Automation Professional** – Automation Anywhere University (2025)
- MongoDB Associate Developer** – MongoDB University (2024)

Education

B.Tech. in Computer Science and Engineering	2027
KL University, Hyderabad	9.95/10
Intermediate (MPC – Maths, Physics, Chemistry)	2022
Sri Chaitanya Junior Kalasala, Hyderabad	96.9%
Secondary School Certificate	2020
Karuna High School	10/10

Achievements

- **Design Patent (2024)** – “Smart Shopping Cart: Automatic Billing System”.
- **Academic Excellence Award (2025)** – Ranked 1st in the Computer Science department at KL University.